

Boeing 737 MAX Crashes

GMBA Case Study Report (2018-2019)

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1 Key Questions & Case Objectives

This case study addresses three high-order management questions through the lens of **The Boeing Company's** Boeing 737 max crashes. The issue is not only a famous business story; it is a decision problem about how leaders use evidence, protect stakeholders, and act when strategy, incentives, and public trust collide.

1. Strategy & Value Creation

- How can data-driven decision-making create sustainable competitive advantage?
- *Case Application:* The Boeing Company needed to interpret market, operational, financial, and stakeholder signals before choosing whether to defend the existing model or make a more fundamental change.

2. Ethics & Governance

- What are the boundaries of data utilization, and who bears responsibility for its consequences?
- *Case Application:* The case raises questions about accountability, transparency, and whether leaders used data to understand reality or to justify a preferred decision.

3. Execution & Transformation

- How should organizations balance innovation speed with operational risk in digital transformation?
- *Case Application:* The decision required leadership to move quickly while managing brand risk, operating constraints, and unintended consequences.

Debate Proposition: “Should executives face prison for safety decisions, and who failed: Boeing or regulators?”

2 Background & Context

2.1 Industry Landscape and Market Environment

Two 737 MAX crashes killed 346 people and exposed failures in software design, pilot disclosure, regulator interaction, and safety governance. The surrounding industry environment made the decision more difficult because competitors, customers, investors, and regulators were already changing expectations about speed, transparency, and value creation.

2.2 Company's Strategic Position

The Boeing Company entered the case with valuable assets: brand recognition, customer relationships, distribution, data, and managerial attention. Those assets also created inertia. The same strengths that made the company important made it harder to admit that an existing strategy, product, policy, or governance model might need to change.

2.3 Current Data/Technology Infrastructure

The relevant data environment includes customer behavior, operational performance, financial reporting, external market signals, and stakeholder feedback. The central analytical challenge is not simply collecting more data; it is deciding which signals should drive action when short-term metrics conflict with trust, safety, or long-term strategic resilience.

3 Core Issues

3.1 Business Problem Definition

The core business problem is how The Boeing Company should respond when evidence suggests that the current approach is producing strategic, ethical, or operational risk. The case connects directly to Engineering ethics, regulatory failure, shareholder vs. stakeholder, but the management dilemma is broader: leaders must decide whether the problem is a temporary execution failure or a sign that the underlying model needs redesign.

3.2 Why a Decision Is Needed Now

Delay can make the problem more expensive. Customers may leave, regulators may intervene, employees may lose confidence, and competitors may use the moment to reposition themselves.

A decision is needed because the organization cannot preserve credibility while treating the issue as routine.

3.3 Constraints

Category	Constraint
Financial	Management must protect cash flow, margins, and investor confidence while making decisions under public scrutiny.
Operational	The organization must align product, process, technology, and data systems fast enough to prevent the issue from compounding.
Brand	The decision affects trust in a highly visible brand where customers, regulators, employees, and partners can react quickly.
Human	Employees, customers, communities, and counterparties bear consequences that cannot be reduced to short-term financial metrics.

4 Data Science / Big Data Perspectives

4.1 Data Landscape

4.1.1 Available Data Types and Sources

- **Customer and stakeholder data:** Complaints, churn, usage, sentiment, willingness to pay, adoption, and support interactions.
- **Operational data:** Defects, cycle time, utilization, incident reports, capacity, inventory, service levels, and process exceptions.
- **Financial data:** Revenue, margin, cash flow, lifetime value, customer-acquisition cost, restructuring cost, and litigation or regulatory exposure.
- **External data:** Competitor moves, media coverage, court or regulatory records, macroeconomic conditions, and industry benchmarks.

4.1.2 Data Quality and Limitations

The main limitation is interpretation. Data can show what happened, but managers still need to determine whether the signal is temporary noise, a structural change, or evidence of deeper governance failure. Historical data may be especially misleading when technology, regulation, or customer expectations are shifting.

4.1.3 Data Governance Issues

The case requires shared definitions of risk, value, and accountability. Without governance, teams can optimize local metrics while creating enterprise-level harm. Data governance should clarify who owns the metric, who can challenge it, and how decisions are documented.

4.2 Analytical Approaches

4.2.1 Applicable Techniques

Technique	Application at The Boeing Company
Cohort and Segment Analysis	Separate customers, products, regions, or stakeholders to identify where value, risk, or backlash is concentrated.
Scenario Modeling	Compare best-case, base-case, and downside outcomes before committing to irreversible strategic moves.
Early-Warning Dashboards	Track leading indicators such as complaints, churn, defect reports, adoption, utilization, or regulatory signals.
Root-Cause Analysis	Connect observed outcomes to organizational incentives, process design, governance, and data-quality limitations.

4.2.2 Technical Requirements

The organization needs integrated reporting across strategy, finance, operations, legal, product, and customer-facing teams. The most important requirement is a trusted decision dashboard that combines leading indicators with financial consequences and stakeholder-risk signals.

4.2.3 ROI Measurement Methods

- Recovery or protection of revenue, margin, cash flow, and enterprise value.
- Reduction in complaints, defects, incidents, churn, or regulatory exposure.
- Improvement in customer trust, employee confidence, and partner stability.
- Evidence that the chosen response improves long-term strategic positioning rather than only short-term optics.

4.3 Ethical Considerations

4.3.1 Privacy and Data Protection

Customer and employee data should be used only for legitimate business analysis, with appropriate access controls and aggregation. Sensitive data should not be used to shift blame onto individuals when the root cause is organizational design.

4.3.2 Algorithmic Bias

Models can overvalue measurable financial outcomes and undervalue trust, safety, fairness, or long-term brand effects. Leaders should test whether dashboards exclude affected stakeholders simply because their harm is harder to quantify.

4.3.3 Transparency and Explainability

Stakeholders need a clear explanation of what the company knew, what it changed, and how future decisions will be governed. Explainability matters because trust recovery depends on credible reasoning, not only on a favorable metric.

5 Strategic Alternatives

Option	Description	Pros	Cons
Option A: Defend the Current Model	Preserve the existing strategy and address the issue with limited tactical changes.	Avoids disruption and protects short-term continuity.	Risks appearing slow, defensive, or unwilling to confront the underlying problem.
Option B: Disciplined Reset	Make targeted changes to strategy, governance, operations, and metrics while protecting the strongest parts of the brand.	Balances accountability with future value creation.	Requires tradeoffs, internal conflict, and clear communication.
Option C: Radical Break	Abandon the contested model, product, or governance structure and rebuild around a new approach.	Signals seriousness and may restore trust quickly.	Can destroy useful assets and may overcorrect before evidence is complete.

Decision and Analysis: The recommended classroom position is **Option B: Disciplined Reset**. It forces management to acknowledge the problem, preserve useful assets, and redesign incentives or operating routines without assuming that every legacy practice must be abandoned. Students should still test whether the facts of the case justify the more conservative Option A or the more radical Option C.

6 Debate Framework

6.1 Pro Side (Affirmative)

“Should executives face prison for safety decisions, and who failed: Boeing or regulators?”

Key Arguments:

1. **Evidence Should Override Habit:** When data and stakeholder signals indicate risk, management should not defend the old model simply because it was historically successful.
2. **Trust Is an Economic Asset:** Reputation, regulatory legitimacy, and customer confidence affect future cash flows.
3. **Governance Converts Data into Action:** Better dashboards are not enough unless decision rights and accountability are clear.

Supporting Evidence:

- The case illustrates how visible brands can face rapid shifts in stakeholder trust.
- Engineering ethics, regulatory failure, shareholder vs. stakeholder provides a theoretical basis for analyzing the decision.
- Verified sources in the reference section provide case chronology, official data, or scholarly frameworks.

Rebuttals to Anticipated Counterarguments:

- *Counterargument:* The company should avoid overreacting to public pressure. *Rebuttal:* A disciplined reset is not panic; it is a structured response that separates useful assets from harmful practices.

6.2 Con Side (Opposition)

“Management should avoid dramatic changes until evidence proves the business model itself is broken.”

Key Arguments:

1. **Short-Term Pressure Can Distort Strategy:** Media, markets, and regulators may reward symbolic action over durable fixes.
2. **Data Can Be Ambiguous:** Early indicators may not prove that a full strategic reset is necessary.
3. **Overcorrection Has Costs:** Radical changes can destroy capabilities, morale, or customer value.

Supporting Evidence:

- Some crises fade after operational corrections, while others reveal deeper strategic failure.
- Not every unpopular decision is unethical; some unpopular choices create long-term value.
- Strong governance requires proportionality as well as accountability.

Rebuttals to Anticipated Counterarguments:

- *Counterargument:* Waiting increases harm. *Rebuttal:* Management can act immediately on risk controls while still gathering evidence before making irreversible strategic commitments.

7 Pre-Class Preparation Questions

1. What are the critical success factors for data-driven decision-making in this case?
 - Consider data quality, stakeholder visibility, incentive design, and leadership accountability.
2. What additional data would have enabled a better decision?
 - Consider customer behavior, operating costs, risk signals, employee feedback, and external benchmarks.
3. If you were the decision-maker, what would you choose and why?
 - Compare the financial, ethical, operational, and reputational consequences of each option.
4. What is the strongest argument for the opposing side?
 - Identify where your preferred decision could fail or create new risks.

8 References (APA 7th Edition)

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- Reason, J. (1990). *Human error*. Cambridge University Press.
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- Jensen, M. C., & Meckling, W. H. (1976). Theory of the firm: Managerial behavior, agency costs and ownership structure. *Journal of Financial Economics*, 3(4), 305-360. <https://doi.org/10.1016/0304-405X%2876%2990026-X>
 - Relevance: Frames the governance problem created when managers, founders, investors, or boards have misaligned incentives.
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 - Relevance: Supports the analysis of board oversight, control rights, and monitoring failure.
- U.S. Department of Justice. (2021). *Boeing charged with 737 Max fraud conspiracy and agrees to pay over \$2.5 billion*. <https://www.justice.gov/archives/opa/pr/boeing-charged-737-max-fraud-conspiracy-and-agrees-pay-over-25-billion>
 - Relevance: Provides official criminal-resolution facts, including the \$2.5 billion settlement and 346 crash victims.
- U.S. House Committee on Transportation and Infrastructure. (2020). *Final committee report: The design, development & certification of the Boeing 737 MAX*. <https://transportation.house.gov/committee-activity/boeing-737-max-investigation>
 - Relevance: Provides congressional investigation evidence on certification, design, and oversight failures.